

Patent prior art and litigation analytics engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
Sahasra Yeruva (2520080062)
Mahathi. Ch(2520080030)
Team Number: 7
Pavan kumar (2520090079)

Section: 13

Under the guidance of

Dr.Dyuti Banerjee
Assistant Professor, AI & DS



Hyderabad-500090, Telangana, India.

ABSTRACT

The Patent Prior Art and Litigation Analytics Engine is a data structures and algorithms based system designed to assist patent examiners and litigators in efficiently analyzing large collections of patent documents. The system searches patent claims for relevant prior-art references, measures claim similarity, analyzes patent citation networks, identifies influential patents, and ranks potential references based on relevance.

The system uses HashMaps and Tries for efficient patent and keyword searching, while Graphs with BFS and DFS are used to represent and analyze citation relationships. Priority Queues rank relevant patents, and Set Cover and Backtracking algorithms are used to determine a minimum set of prior-art references that collectively cover the required elements of a target claim. An examiner-assignment module also matches patent applications with suitable examiners based on expertise and workload.

By combining these DSA techniques, the proposed engine provides an efficient and structured approach to prior-art discovery, patent relationship analysis, and litigation research, reducing the time and effort involved in manual patent analysis while serving as a decision-support tool rather than a replacement for legal judgment.

Signature of the Guide